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Platform and Ecosystem

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You use these platforms every day – but who really wins?

Who are the stakeholders?

Warm-Up Poll

1. Which group creates most of the value on these platforms?
    A. Platform owners (Tencent, Alibaba, ByteDance)
    B. Merchants / drivers / creators
    C. Users
    D. Algorithms & data
    E. Regulators

Value Creation

vs

Value Capture/Distribution

2. Who captures most of the value?

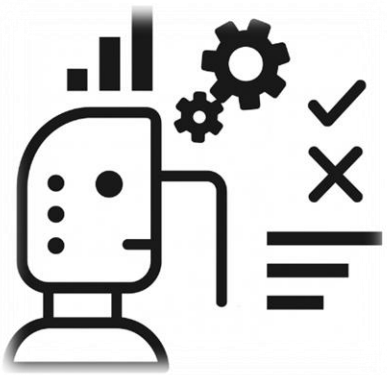
Discussion: A Meituan Driver’s Perspective

- Scenario: A Meituan delivery driver says
  - Without Meituan, I have no orders.
  - With Meituan, I have no bargaining power.
- Discussion
  - Is Meituan empowering or exploiting drivers?
  - Who sets prices, rankings, and incentives?
  - Is this a technology problem, or a governance choice?

Platforms don’t just connect people
— they decide the rules of the game.

Platform Power: Invisible but Decisive

- The most powerful manager you never meet
  - Algorithms decide visibility
  - Rankings shape income
  - Rules change unilaterally
  - Appeals are limited or automated



- Question: Who designed these rules — engineers or strategists?

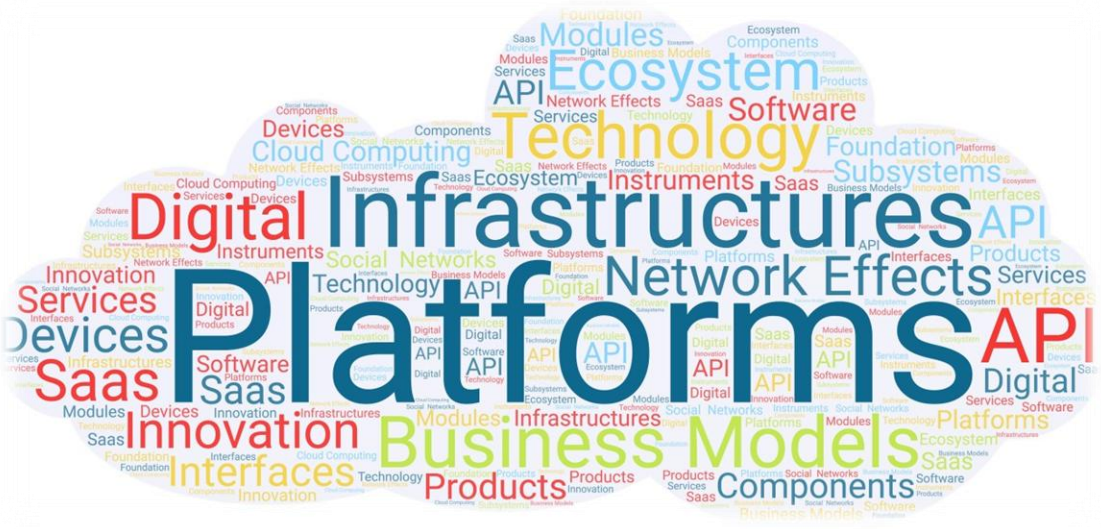
Is “More Users” Always Better?

- When platforms start to hurt their own ecosystem?
- Examples
  - Douyin content saturation → creator burnout
  - Taobao merchants racing to the bottom on price
  - Didi surge pricing → user backlash
- Can a platform grow too fast?
- What matters more for platform success: technology, users, data, or strategy?

From Intuition to Strategy

So what exactly is a platform — strategically?

- Platforms are not just apps or marketplaces.
- They are **organizational forms** enabled by information systems.



Platform: A Strategic Definition

- A platform is a organizational form that
  - Enables direct interactions between two or more distinct user groups
  - Coordinates interactions through rules, algorithms, and data
  - Information systems are the core infrastructure
- Key emphasis
  - Direct interactions
  - Multiple sides
  - IS-enabled coordination

Platforms ≠ Digital Pipeline

Pipeline	Platform
Produces value internally	Orchestrates value externally
Optimizes internal processes	Optimizes interactions
Controls via hierarchy	Controls via rules & IS
IT supports operations	IS is the business

Value Creation vs Value Capture

- Value is co-created by users & complementors.
- Platforms design how value is distributed.
- IS determines
  - Visibility
  - Matching quality
  - Trust
  - Monetization

What Must Exist for a Platform to Function

Four essential components

- Distinct user groups
- Interaction rules
- Matching mechanism
- Trust infrastructure

If any one of these fails, what happens?

IS as the Coordination Engine

Map IS → Platform Functions

- Databases → user profiles, history
- Algorithms → matching, ranking, pricing
- Interfaces → user behavior shaping
- Analytics → learning and optimization

Platforms scale because software scales decision-making

Not All Platforms Compete the Same Way

Why Classification Matters

- Platform strategy depends on what kind of value is being orchestrated
- Different platforms require different IS capabilities, incentives, and governance

- Transaction platforms
  - Examples: Meituan, Didi
  - Focus: efficiency & liquidity
- Innovation platforms
  - Example: WeChat mini-programs
  - Focus: third-party innovation
- Hybrid platforms
  - Examples: Alibaba, Tencent
  - Focus: ecosystem dominance

Platform vs Ecosystem: Core vs Expansion

Platform = the Core Engine

- A platform is the technical and rule-based core.
- It defines:
  - Who can participate
  - How interactions occur
  - How value is measured and allocated
- Built and controlled primarily through information systems

Ecosystem = the Growth Layer

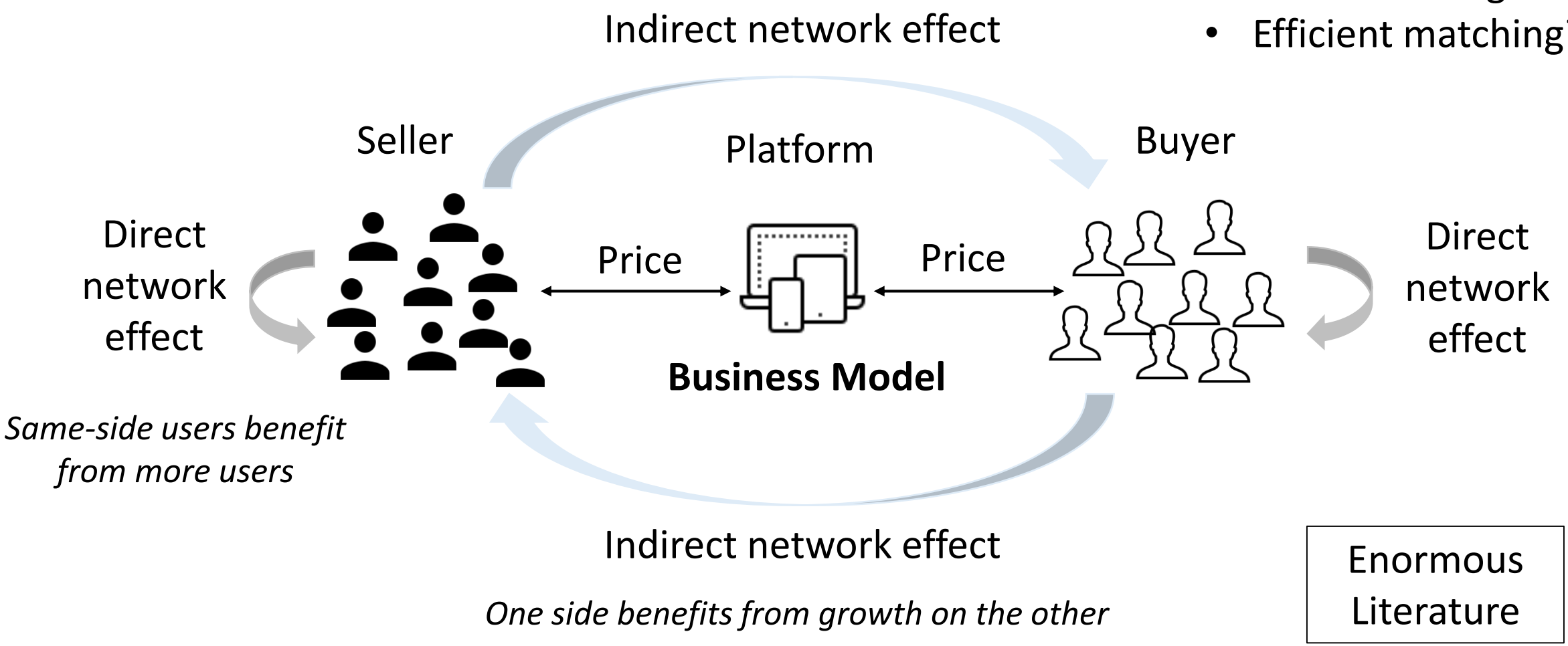
- An ecosystem is the network of external actors around the platform:
  - Complementors (merchants, developers, creators)
  - Partners and service providers
  - Sometimes regulators and standards bodies
- Ecosystems expand:
  - Variety
  - Innovation speed
  - Market reach

More Users ≠ Better Platform

- Congestion increases
  - Too many sellers → lower visibility
  - Too many drivers → idle time & income drop
  - Too many creators → attention fragmentation
- Quality deteriorates
  - Low-quality merchants / content flood the platform
  - Trust mechanisms struggle to scale
  - Users face higher search costs
- Incentives misalign
  - Race to the bottom on price
  - Creators chase algorithms, not value
  - Complementors lose motivation to innovate
- Governance lags behind scale
  - Rules are unclear or change too often
  - Appeals become automated and opaque
  - Perceived unfairness increases churn

Network Effect

- Number?
- Data?
- Positive vs negative?
- Efficient matching?



Enormous Literature

Network Effects Do Not Guarantee Monopoly

- Network effects may lead to:
  - Winner-take-all
  - Winner-take-most
  - Coexistence
- What determines the outcome
  - Multi-homing (can users join multiple platforms?)
  - Switching costs
  - Differentiation
  - Regulation

Network effects create advantage — strategy decides the market structure.

Growth Can Destroy Value

- Network effects become negative when
  - Congestion increases
  - Quality deteriorates
  - Trust erodes
  - Attention becomes scarce
- Any example?

Growth without governance leads to negative network effects.

Network Effects Are Managed, Not Automatic

1. Subsidize One Side
   - Often subsidize the demand side first
   - Goal: create liquidity and participation
   - Example: free rides, coupons, free content
2. Control Entry and Quality
   - Who is allowed to join
   - How quality is verified or ranked
   - Prevents congestion and trust erosion
3. Design Incentives
   - Pricing rules
   - Revenue sharing
   - Visibility and ranking rewards
4. Use Algorithms to Balance Sides
   - Matching supply and demand
   - Preventing oversupply or undersupply
   - Dynamically adjusting outcomes

Why Network Effects Are Powerful — and Fragile

- Why Network Effects Create Strong Advantage
  - More users → more interactions → more data
  - Data improves matching, ranking, and personalization
  - Users develop habits → higher switching costs
  - 👉 Result: Self-reinforcing growth

Network effects create advantage — governance determines whether it lasts.

- Why Network Effects Are Fragile
  - Advantage depends on user trust, perceived fairness, ecosystem participation
  - These can break due to poor rule changes, algorithm shocks, regulatory intervention
  - 👉 Network effects are not permanent assets

Network Effects: What Managers Must Understand

- Network effects explain platform power
  - Platforms scale by adding participants, not assets
  - Advantage comes from demand-side feedback loops
- Growth alone is not a strategy
  - More users ≠ more value
  - Without design, growth creates congestion and conflict
- Network effects can turn negative
  - Quality dilution, trust erosion, ecosystem disengagement
- Governance determines durability
  - Rules, incentives, and algorithms shape outcomes

If you were the platform owner:

- When would you slow growth?
- Whose interests would you protect first?
- What signals would tell you network effects are weakening?

Why Power, Not Technology, Decides Platform Outcomes

- Platforms do more than enable transactions
- They define the rules of participation
- These rules shape:
  - Value creation
  - Value capture
  - Long-term survival
- Key insight
  - Technology enables scale —governance determines who benefits from it.

Think: If two platforms use similar technology, why does one dominate?

Platform Governance: A Strategic View

- Platform governance includes:
  - Participation rules (who can join, who cannot)
  - Value rules (fees, commissions, revenue sharing)
  - Control mechanisms (ranking, matching, moderation)
  - Dispute resolution (appeals, enforcement)
- Important clarification
  - Governance is embedded in code, algorithms, and data flows
  - Not just written policies
- Most users never read platform rules — they experience them.

Where Platform Power Comes From

Platforms gain power by controlling:

- Access
  - Approval, removal, suspension
- Visibility
  - Ranking, recommendations, exposure
- Monetization
  - Pricing rules, commissions, fees
- Data
  - Who sees what
  - Who learns faster

Which of these would you least want a platform to control over you?

Openness vs Control: The Central Governance Trade-Off

- Why platforms open up
  - Attract users and partners faster
  - Encourage innovation and variety
  - Accelerate network effects
- Why platforms tighten control
  - Maintain quality and trust
  - Reduce coordination and compliance costs
  - Protect core business and data
- What changes over time
  - Early stage → openness matters more
  - Growth stage → balance becomes critical
  - Mature stage → control often increases
  - Governance is dynamic, not a one-time decision.

Examples

- Apple App Store: open entry → strict review
- Alibaba: merchant growth → governance tightening
- Douyin: creator expansion → algorithmic control

Conflict Is Inevitable

- Platforms sit in the middle of competing interests
  - Users want low prices and convenience
  - Complementors want income stability and fairness
  - Platforms want growth, data, and control
  - 👉 These goals cannot be fully aligned.
- Common triggers of conflict
  - Rule changes without consultation
  - Fee or commission increases
  - Algorithm updates affecting visibility or income
  - Platforms entering complementors' markets

Conflict is not a failure of platforms — it is a consequence of platform power.

Algorithmic Power

- Algorithms quietly decide:
  - Visibility — who gets seen
  - Access — who gets matched or excluded
  - Rewards — income, traffic, growth
  - Discipline — demotion, throttling, removal
- Algorithmic power is:
  - Continuous (every interaction)
  - Scalable (millions of decisions instantly)
  - Opaque (rules are hidden or simplified)
  - Unilateral (appeals are limited or automated)

Algorithms feel neutral —

but they reflect strategic choices

about power and value.

When Platform Power Triggers Pushback

- Pushback occurs when platforms are seen as:
  - Too powerful, opaque, unfair, difficult to exit
- Sources of Pushback
  - Users: loss of trust, migration to alternatives
  - Complementors: protests, exit, collective action
  - Governments: antitrust enforcement, data & algorithm regulation
- Typical Regulatory Focus
  - Market dominance and competition
  - Algorithmic transparency
  - Data usage and protection
  - Platform responsibility

Think: Should governments regulate *platform behavior* or *platform power*?

- Outcomes vs processes
- Innovation vs fairness

Summary

- Platforms are strategic IS architectures
  - Value is created through designed interactions
  - Information systems are the core business logic
- Network effects drive scale — but not automatically
  - Growth can create or destroy value
  - Advantage depends on how participation is shaped
- Governance determines power
  - Platforms rule through algorithms, rules, and data
  - Power is exercised quietly — conflict appears later
- Power invites pushback
  - Users, partners, and regulators respond to unfair governance
  - Sustainable platforms require legitimacy, not just dominance